



# ELAN

## *A GIS decision-support tool for integrated urban water management by nature-based solutions*

Date : 25-09-2025

Presentation : Sophie Aubier



*ceci n'est pas une conférence*

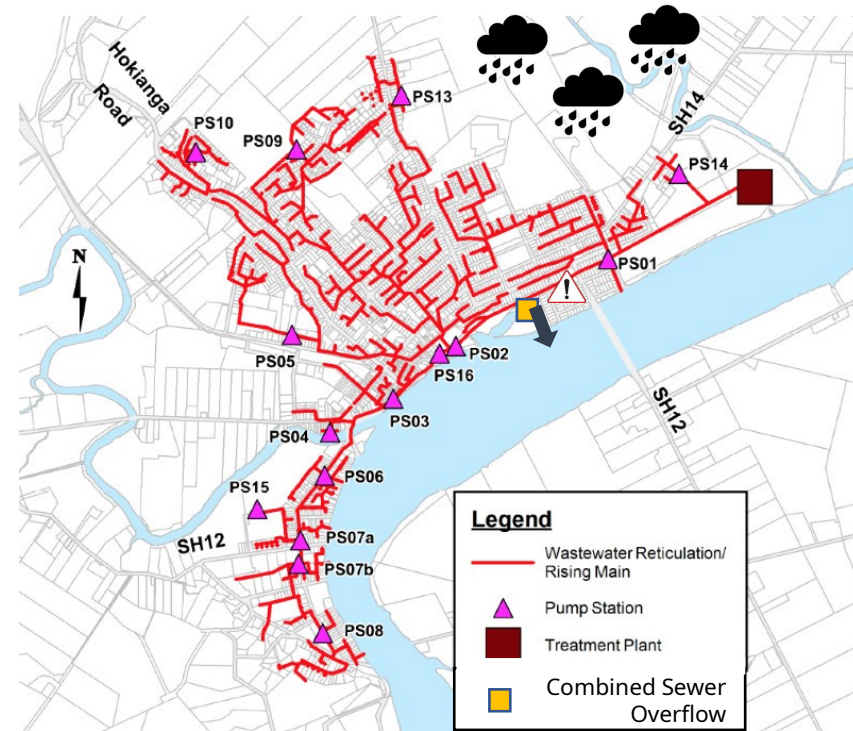


# A QGIS plugin to address two questions

How to connect an area with no existing sanitation network ?  
(centralised / decentralised management)

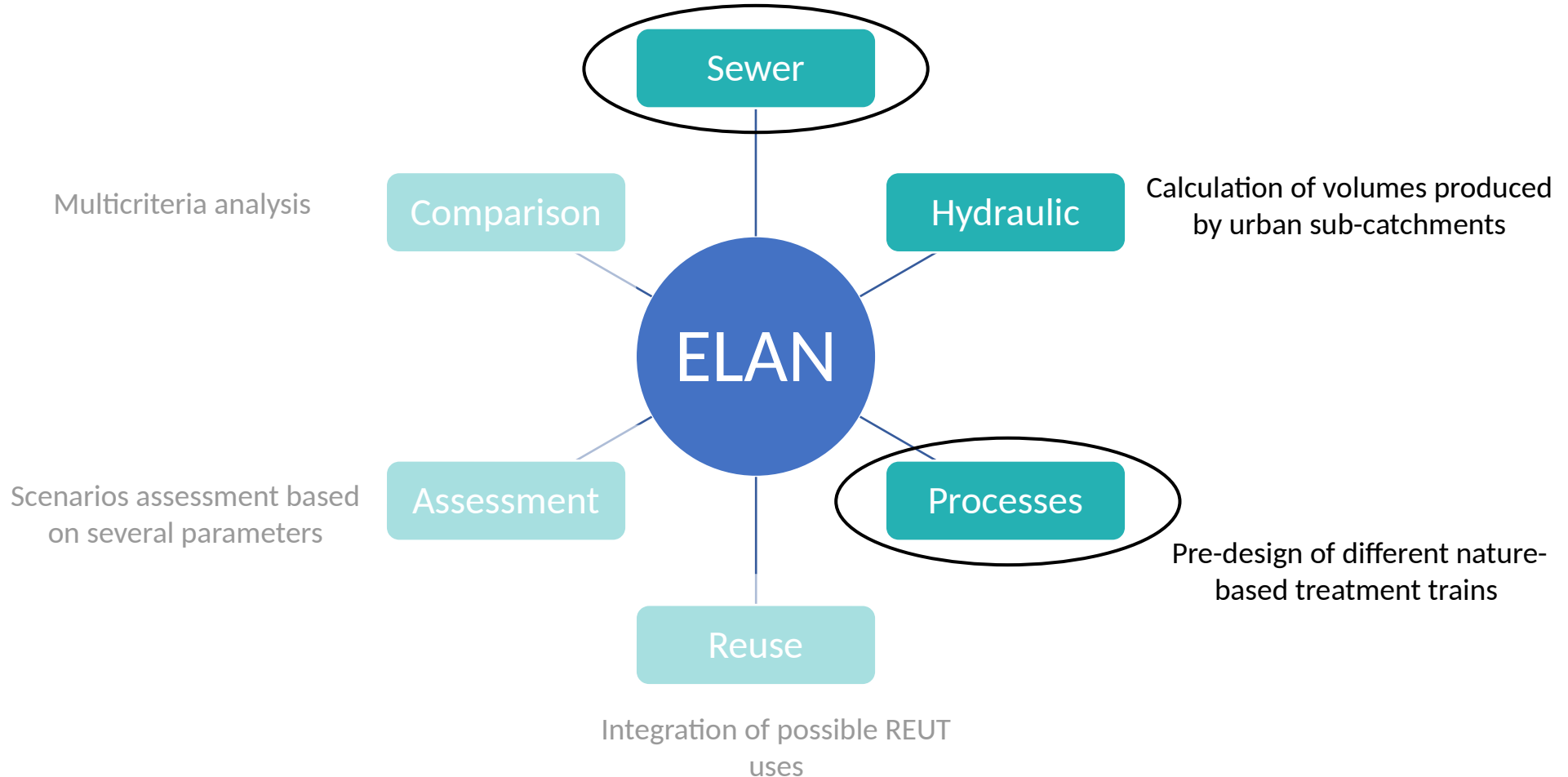


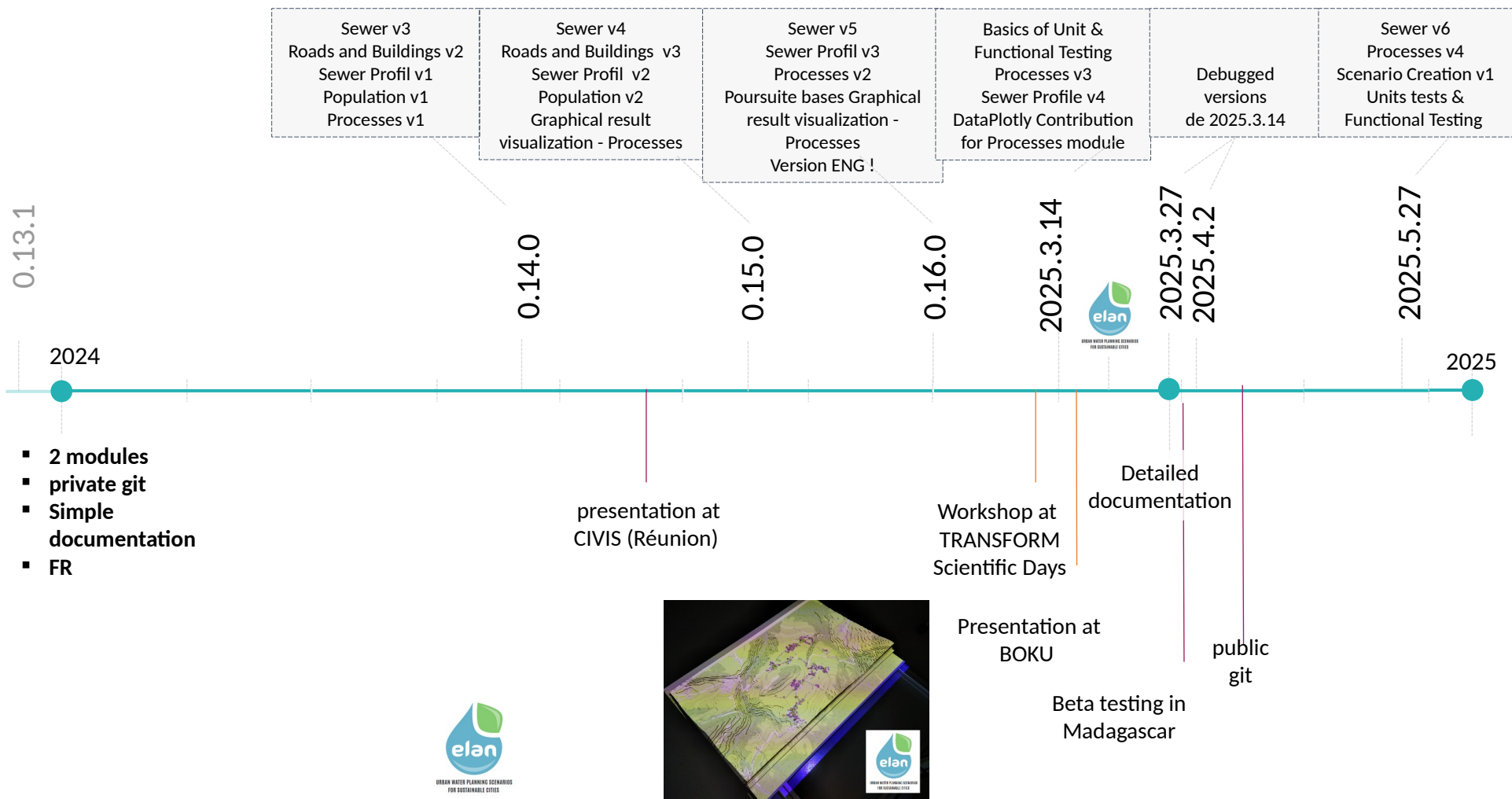
How to limit existing storm water overflows ?  
(wastewater and storm water)



# A modular structure:

Pre-design of a 3D sanitation network (WW)





# A modular structure:



To prepare the data



To design sewer networks and explore the results



To investigate the preliminary sizing proposed by the Network module outputs

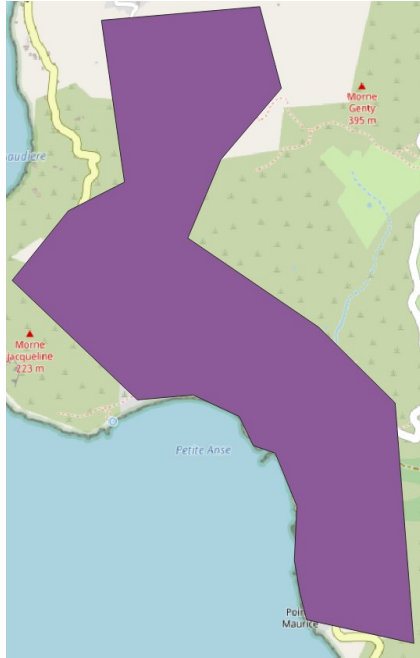


To create scenarios, evaluate them, and compare



To explore and make an informed decision

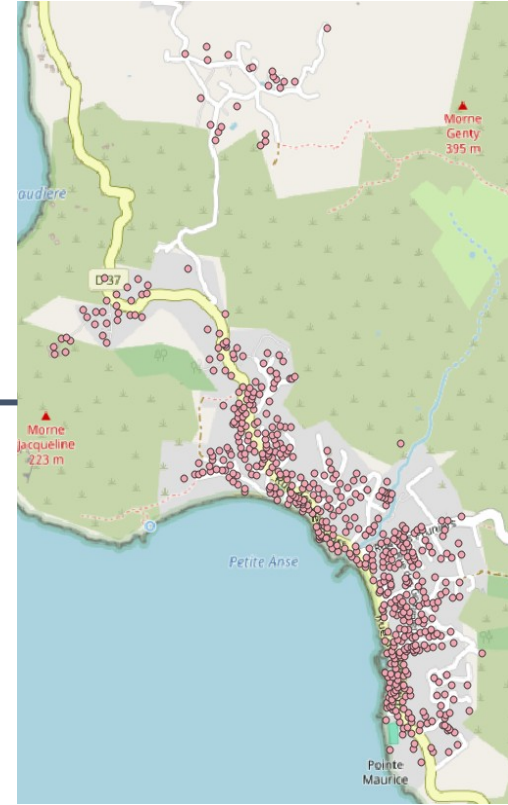
# To prepare the data:



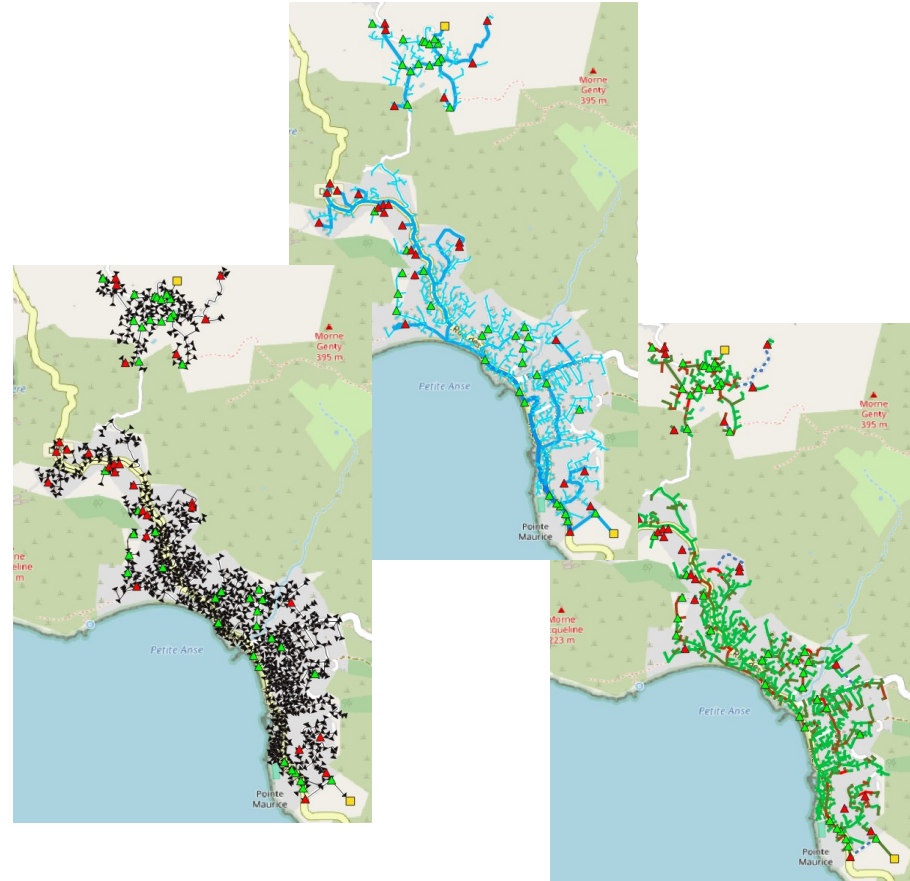
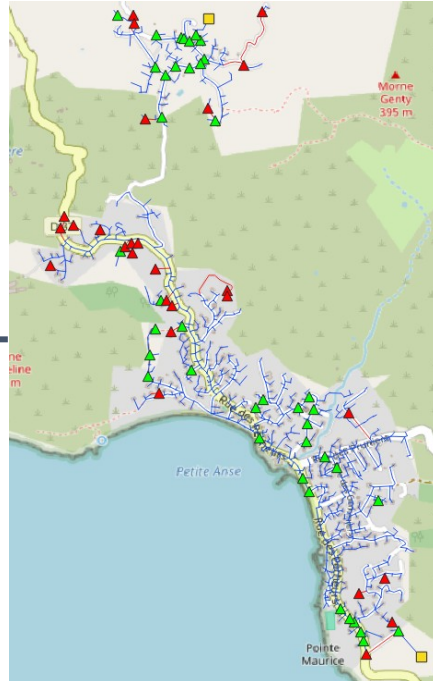
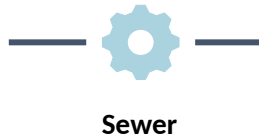
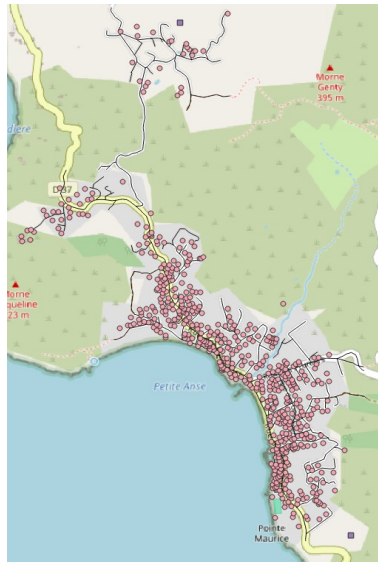
  
**Roads  
and Buildings**



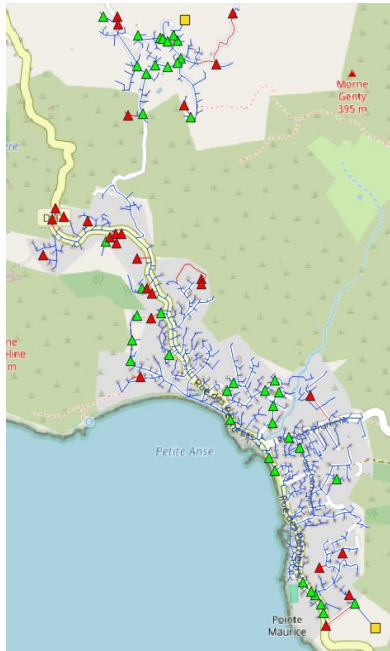
  
**Population**



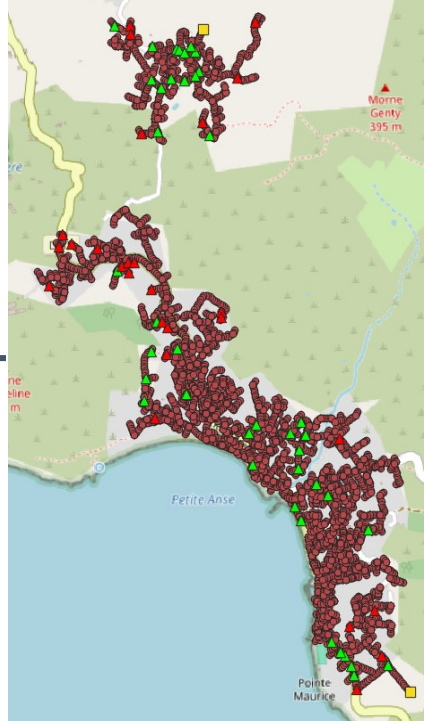
# To design sewer networks and explore the results:



# To investigate the Network module outputs

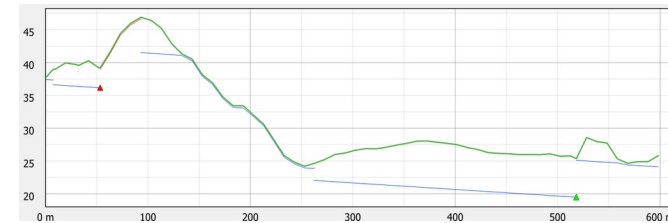
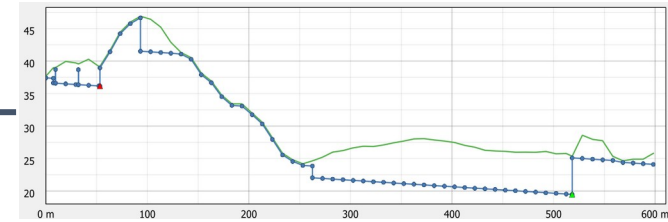
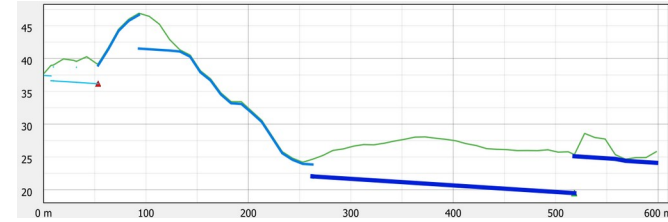


Sewer Profile

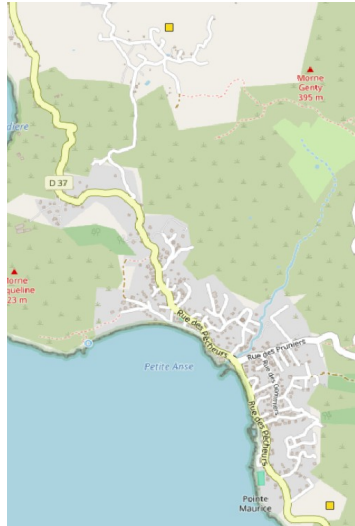


QGIS

Select the sewer  
pipes ->  
Elevation Profile



# Creating Scenarios: Evaluation & Comparison

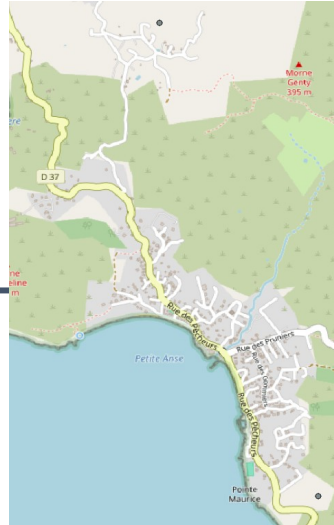


Processes  
Tropical or  
Temperate  
climate

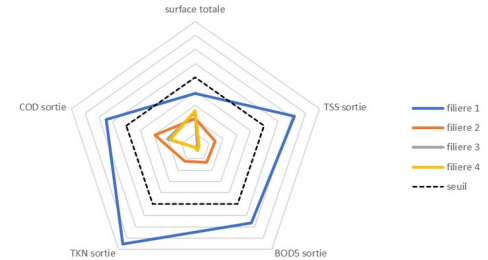
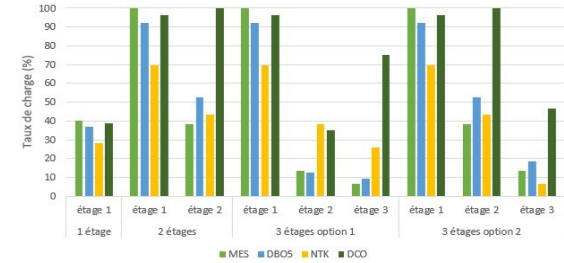
STEU — Total des entités: 2, Filtrées: 2, Sélectionnées: 0

|   | niveau rejet MES | niveau rejet DBOS | niveau rejet NTK | niveau rejet DCO | niveau rejet NT |
|---|------------------|-------------------|------------------|------------------|-----------------|
| 1 | 35               | 35                | 20               | 125              | NULL            |
| 2 | 35               | 35                | NULL             | 125              | NULL            |

Montrer toutes les entités



QGIS graphical  
vizualisation

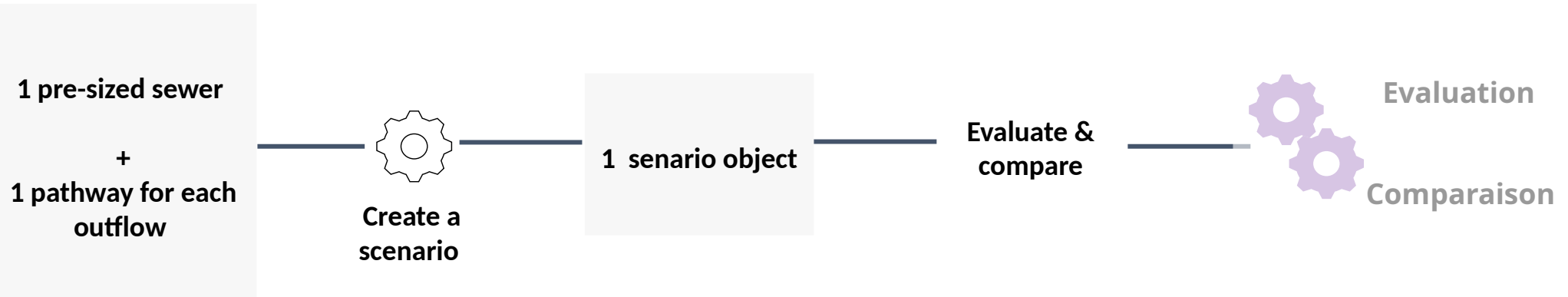


Couche de filières — Total des entités: 12, Filtrées: 12, Sélectionnées: 0

| id filière | descriptif filière | coordonnées gps                       | taux de charge MES par étape de traitement | taux de charge DBOS par étape de traitement | taux de charge NTK par étape de traitement |
|------------|--------------------|---------------------------------------|--------------------------------------------|---------------------------------------------|--------------------------------------------|
| 0          | VdNS1-VdNS2        | (707878.2273226, [100.0, 38.2]        | [92.01, 52.72]                             | [69.79, 43.36]                              |                                            |
| 1          | VdNS1-VdNS2-VdNS2  | (707878.2273226, [100.0, 38.2, 13.44] | [92.01, 52.72, 18.55]                      | [69.79, 43.36, 6.7]                         |                                            |
| 2          | VdNS5-VdNS2        | (707878.2273226, [100.0, 67.2]        | [92.01, 92.75]                             | [69.79, 76.29]                              |                                            |
| 3          | VdNS5-VdNS2-VdNS2  | (707878.2273226, [100.0, 67.2, 13.44] | [92.01, 92.75, 18.55]                      | [69.79, 76.29, 14.53]                       |                                            |
| 4          | VdNS1              | (707878.2273226, [40.1]               | [36.9]                                     | [27.99]                                     |                                            |
| 5          | VdNS5              | (707878.2273226, [40.1]               | [36.9]                                     | [27.99]                                     |                                            |
| 7          | 0                  | VdNS1                                 | (708750.1184972, [100.0]                   | [92.01]                                     | [69.79]                                    |
| 8          | 1                  | VdNS1-VdNS2                           | (708750.1184972, [100.0, 38.2]             | [92.01, 52.72]                              | [69.79, 43.36]                             |
| 9          | 2                  | VdNS5                                 | (708750.1184972, [100.0]                   | [92.01]                                     | [69.79]                                    |
| 10         | 3                  | VdNS1-VdNS2-VdNS2                     | (708750.1184972, [100.0, 38.2, 13.44]      | [92.01, 52.72, 18.55]                       | [69.79, 43.36, 6.7]                        |
| 11         | 4                  | VdNS5-VdNS2                           | (708750.1184972, [100.0, 67.2]             | [92.01, 92.75]                              | [69.79, 76.29]                             |
| 12         | 5                  | VdNS5-VdNS2-VdNS2                     | (708750.1184972, [100.0, 67.2, 13.44]      | [92.01, 92.75, 18.55]                       | [69.79, 76.29, 14.53]                      |

Montrer toutes les entités

# To explore and make an informed decision



 Addresses a professional/sector-specific challenge

 Creates a custom open-source solution

 Leverages existing QGIS tools when possible:

- Contribution to DataPlotly
- Contribution to Pysewer
- Collaboration with UFZ

# Ressources :

Git : <https://gitlab.com/elan7835313/elan>

Documentation : <https://elan7835313.gitlab.io/elan/>

Project page : <https://reversaal.lyon-grenoble.hub.inrae.fr/projets/digital/elan>

